Directory Systems

1 Basic Idea

Definition

A **directory** is a special file used by the file system to keep track of file names and organize files.

- File systems need a way to locate and organize files.
- Directories, also called folders, provide this organization.
- A directory may contain file names.
- In modern systems, a directory may also contain other directories.
- Directory organization affects how easily users can find and group files.

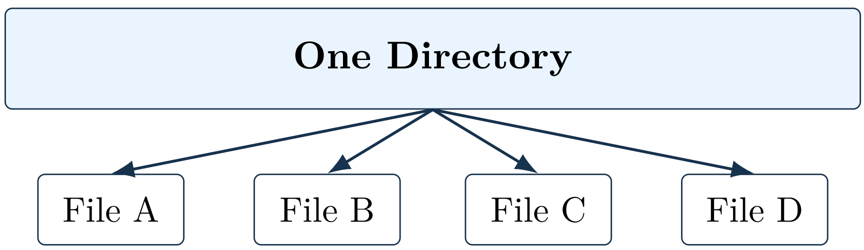
Core Point

Directories are themselves files, but their contents are interpreted by the file system as file-system organization data.

2 Single-Level Directory System

Single-Level Directory

A **single-level directory system** has one directory containing all files in the system.



- There is only one directory.
- Every file is listed in that directory.
- Since there is only one directory, the term root directory is not very important.
- Early personal computers often used this design.
- The CDC 6600 supercomputer also used a single directory, even with many users.
- The main reason was software simplicity.

3 Advantages of Single-Level Directories

- Very simple to implement.
- Very simple to understand.
- File lookup is straightforward.
- There is only one place to search.
- Still useful for very small systems.
- Sometimes used in embedded devices, digital cameras, or simple music players.

4 Limitations of Single-Level Directories

Main Problem

A single directory becomes hard to use when there are many files or many users.

- All files share one namespace.
- File-name conflicts are more likely.
- Related files cannot be grouped naturally.
- Finding a file becomes difficult when thousands of files exist.
- It does not match how modern users organize work.

5 Need for Hierarchy

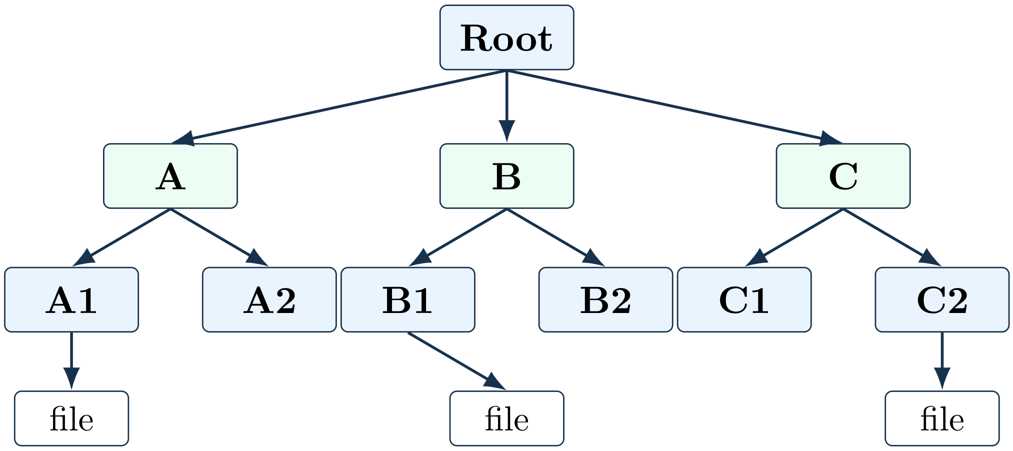
Motivation

Users need to group related files together, especially when they have many projects or many categories of work.

- A user may have files for a book.
- The same user may also have student programs.
- Other files may belong to compiler projects, grant proposals, email, meeting minutes, papers, or games.
- Keeping all of these in one directory would be confusing.
- A hierarchy lets each group have its own directory.

6 Hierarchical Directory System

> **Hierarchical Directory**
>
> A **hierarchical directory system** organizes directories as a tree, with directories containing files and subdirectories.



- The directory structure forms a tree.
- The root directory is at the top.
- Directories below the root can belong to different users.
- Each user can create private subdirectories.
- Subdirectories can represent projects, courses, documents, or any other grouping.
- Files are placed inside the relevant directory.

7 Benefits of Hierarchical Directories

- Users can create as many directories as needed.
- Related files can be grouped together.
- File-name conflicts are reduced because names are local to directories.
- Multiple users can share one file server while keeping separate directory trees.
- Large collections of files become easier to navigate.
- This structure is used by nearly all modern file systems.

> **Modern Design**
>
> Hierarchical directories are standard in modern systems because they scale from a few files to millions of files across many users.

8 Single-Level vs. Hierarchical

Feature	Single-level directory	Hierarchical directory
Structure	One directory contains every file.	Directories form a tree.
Simplicity	Very simple.	More complex but more useful.
Grouping	No natural grouping.	Related files can be grouped in subdirectories.
Naming conflicts	More likely.	Less likely because directories provide separate contexts.
Best use	Very small or embedded systems.	Modern multiuser and general-purpose systems.

9 Key Contrast

- A single-level directory is easy to implement but does not scale well.
- A hierarchical directory is more complex but much better for organization.
- Early systems used single directories because storage and software were simpler.
- Modern systems use directory trees because users manage many files and many projects.

Final Point

Directories organize file names. Single-level directories are simple, but hierarchical directories provide the structure needed for modern file systems.